DonProof FB-S

High-performance, fully bonded PVC membrane extruded with swelling non-woven for below-grade waterproofing

DESCRIPTION

DonProof FB-S is a high-performance, pre-applied waterproofing membrane system composed of a PVC sealing layer co-extruded with an advanced swelling non-woven geotextile. This innovative non-woven layer incorporates water-reactive polymers that swell upon contact with moisture, providing self-sealing and self-healing capabilities.

DonProof FB-S is installed before the steel reinforcement is fixed and the concrete is poured. The unique configuration of the membrane enables a full mechanical bond with freshly poured concrete, eliminating water migration between the membrane and the structure.

DonProof FB-S technology offers superior resistance to water, gases (e.g., radon), salts, and chemical exposure, making it ideal for demanding below-grade applications.

APPLICATIONS

DonProof FB-S can be used in both pre-applied and post-applied configurations and is particularly suited for the following applications:

- Underground waterproofing applications, including foundations and basements, retaining walls and floor slabs, underground car parks and garages.
- Sealing of precast concrete elements to enhance joint tightness and overall durability.
- Protection of concrete structures exposed to aggressive groundwater or other chemically active environments.
- Marine and salt-contaminated environments.
- Concrete structures in chemically.
- aggressive environments and ground water.

ADVANTAGES

- Fully bonded system, forms a permanent mechanical bond with freshly poured concrete.
- Can be used both pre- and post-applied.
- Swelling fleece layer offers self-healing properties, maintaining integrity even after minor damage or post-installation movement
- Suitable were durable protection against groundwater ingress and harmful contaminants.
- Excellent mechanical properties, tough, and durable.
- Easy to handle and install on-site.
- Excellent chemical resistance, can be used in saline environments.
- Excellent bond and shear strength ensuring long-lasting adhesion to concrete.
- Acts as an external barrier to protect floor slabs and exterior walls from soil moisture and groundwater.
- Effective barrier against radon gas infiltration.
- Provides high flexibility and crack-bridging capacity through mechanical bonding with freshly poured concrete.
- Compatible with bitumen.

TECHNICAL PROPERTIES:

Material:	PVC-P membrane + PP Swelling nonwoven
Visible defects: EN 1850-2	No visible defects (Pass)
Straightness: EN 1848-2	75 mm/10 m (Pass)
Working temperature:	-5 to 50°C
Thickness: EN 1849-2	1.6 mm ± 10%
Weight EN 1849-2	1.6 ± 10% kg/m²
Tensile strength: (MD/CMD) EN ISO 12311-2 (A)	≥ 980 N/50 mm
Elongation: (MD/CMD) EN ISO 12311-2 (A)	≥ 240%
Water tightness: ASTM D 5385	≥ 690 kPa
Water tightness: EN 1928 (B)	≥ 60 kPa @ 24 hr ≥ 500 kPa @ 72 hr
Watertightness in case of subsequent cracks: ASTM D 5385	Mechanical bond with freshly poured concrete (tight) 6.9 bar/≥ 3.2 mm
Impact resistance: EN 12691 (A)	≥ 600 mm
Static load resistance: EN 12730 (A) EN 12730 (B)	≥ 20 kg/24 hr (Pass)
Tear resistance: (MD/CMD) EN 12310-1	≥ 500 N (Nail shank)
Joint shear resistance: EN 12317-2	≥ 300 N/50 mm Collapse outside of the overlapping

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STANDARDS

DonProof FB-S complies with DIN EN 13967 Type A and type T waterproofing membrane with moisture barrier and groundwater barrier, and BS 8102 combining Type A (barrier protection) and Type B (integral protection) systems as per the technical properties table.

METHOD OF USE

DonProof FB-S must be installed by experienced and qualified personnel.

SURFACE PREPARATION

Ensure all substrates are solid, clean, and free of debris, sharp edges, or voids greater than 12 mm. Lean-mix concrete or compacted sands are suitable base substrates.

All upstands, corners, penetrations, and interfaces must be properly detailed before installing the waterproofing membrane. A 20 mm fillet should be formed at the base of the retaining walls, formwork stop ends, and upstands using a modified cementitious mortar.

The membrane is designed to bond directly with fresh reinforced concrete; thus, surface preparation must follow EN 1992 design standards.

SEPARATION/PROTECTION LAYERS BELOW THE MEMBRANE

A cushion layer of geotextile membrane, weighing no less than 300g/m², shall be installed directly over the blinding concrete as part of the preparation process.

Prior to placement, the substrate should be thoroughly cleaned to ensure it is free from any loose materials, sharp projections, or debris that could damage the membrane.

APPLICATION

DonProof FB-S membrane should be applied to a properly prepared substrate, with the fleece facing the concrete that is to be waterproofed.

The face of the selvedge on the first roll should be cleaned using a clean, dry cloth. The subsequent roll should then be positioned so that the edge selvedge of the previous roll is overlapped by a minimum of 50 mm ensuring accurate alignment and consistent overlap.

All overlaps are to be sealed using heat-welding equipment and should be compressed with a wooden roller.

TECHNICAL PROPERTIES:

Crack bridging ability: ASTM D 5385	≥ 3.2 mm (Pass)
Durability against artificial aging: EN 1928 (B)	12 weeks @ 70°C 60 kPa/24 hr (Pass)
Durability against chemicals: $Ca(OH)_2/H_2SO_3/NaCl$ EN 1847 EN 1928 (B)	28 days @ 23°C 60 kPa @ 24 hr 500 kPa @ 72 hr (Pass)
Compatible with bitumen: EN 1548 EN 1928 (B)	60 kPa @ 24 hr 500 kPa @ 72 hr (Pass)
Root resistance: DIN CEN/TS 14416	Membrane and overlapping (Passed)
Reaction to fire: EN 13501-1 EN ISO 11925-2	Class E

The ends of the membrane must be staggered by a minimum of 300 mm. To achieve this, the ends of the subsequent roll should be butted tightly together, ensuring that no gaps greater than 12 mm are present between roll ends.

150 mm wide strip of DonProof FB-S should be placed over the center of the butt joint using an approved adhesive and compressed using a wooden roller to ensure complete adhesion.

The installed membrane must be inspected for any damage prior to the placement of concrete. If any damage is identified, it should be repaired by placing a patch of the membrane over the affected area and bonding it with heat-welding.

If cleaning is required, the membrane may be cleaned using a soft brush or a low-pressure system with cold water. Any standing water or construction debris must be removed from the membrane before the concrete is placed.

REMARKS

- Concrete must be of a high quality, fully compacted, properly placed and properly cured.
- Care should be taken when using vibrating equipment to prevent damage to the installed waterproofing system.
- A minimum overlap of 50 mm should be maintained at the edges of the membrane, and roll ends should be staggered by at least 300 mm.

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PACKAGING

Thickness	1.6 mm
Width	2.2 m + 1%
Length	25 m
Roll	85.5 kg
Pallet	7 rolls/pallet

STORAGE

DonProof FB-S has a shelf life of 12 months from the date of manufacturer if stored in their original packaging at temperatures between 5°C and 30°C, and kept dry and protected from direct sunlight, snow, ice, water, heat or heat sources.

Do not stack any sharp items or additional pallets on top the DonProof FB-S membrane during transport and storage.

If these conditions are exceeded, contact DCP Technical Department for advice.

CAUTIONS

HEALTH AND SAFETY

DonProof FB-S is not directly associated with anyknown hazard during normal use.

For further information, refer to the Material Safety Data Sheet.

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A wide range of construction chemical products are manufactured by DCP which include:

- Concrete admixtures.
- Surface treatments
- Grouts and anchors.
- Concrete repair.
- Flooring systems.
- Protective coatings.
- Sealants.
- Waterproofing.
- Adhesives.
- Tile adhesives and grouts.
- Building products.
- Structural strengthening.

Note:
We endeavour to ensure that any information, advice or recommendation we may give in product literature is accurate and correct. However, because we have no control over where and how products are applied, we cannot accept any liability arising from the use of the products.